



SAFETY DATA SHEET

SILICON METAL POWDER

C.A.S. Number: 7440-21-3

SECTION 1 – IDENTIFICATION

Product/Material: SI-100/101

Product Name: Silicon Metal Powder

CAS No.: 7440-21-3

Trade Name(s): Silicon, Silicon Metal, Micronized Silicon Metal, Silicon Metal Powder

Supplier's Details: Atlantic Equipment Engineers, Inc.
Address: 24 Industrial Avenue, P.O. Box 181
Upper Saddle River, NJ 07458

Telephone/Fax Numbers: (201) 828-9400 / (201) 387-0291

Emergency Contact: CHEMTREC (800) 424-9300
(703) 527-3887 (Outside the USA)

Product Use: Alloy additive

Restrictions: None identified.

Chemical Formula: Si

Appearance: Grey metallic color

Odor: None

SECTION 2 – HAZARDS IDENTIFICATION

GHS Classification

Health: Airborne dust generated through the use or handling of the product may result in respiratory tract and/or eye discomfort.

Physical: Material is not known to be a water reactive flammable solid material based on United Nations recognized analytical methods.

Dust from this product under certain conditions may be combustible in a dust cloud form.

If this product is not used in a process creating dust, the above hazard would not apply, so the product would not be considered hazardous under HAZCOM 2012.

OSHA Hazard Category: Combustible Dust

GHS Category: None

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GHS Pictogram Label(s)

None

Signal Word

Warning

Hazard Statements

May form combustible dust concentrations in air.

Precautionary Statements

- Avoid breathing dust, as it may irritate and dehydrate mucous membranes.
- Keep away from oxidizing agents and halogen acids.
- Airborne dust under certain conditions may be combustible.
- Avoid skin contact – Dust may irritate skin.
- Avoid eye contact – Dust may irritate and lead to dryness of the eyes.
- Do not ingest.

SECTION 3 – COMPOSITION / INFORMATION ON INGREDIENTS

<u>Ingredient</u>	<u>CAS No#</u>	<u>Percent %</u>
Silicon	7440-21-3	98-99.5

SECTION 4 – FIRST-AID MEASURES

Inhalation:	Remove victim to fresh clean air.
Ingestion:	Seek immediate medical attention.
Skin Contact:	Wash skin with soap and water.
Eye Contact:	Flush eyes with water for 15 minutes.

**Seek immediate medical attention if any symptoms persist*

SECTION 5 – FIRE-FIGHTING MEASURES

Dry silicon metal in the form of lumps or granules (bottom screen > 200 mesh [0.074 mm]) is not combustible.

Extinguishing Media: Dry powder, dry sand, or CO₂

Special Information: Silicon metal airborne dust may under certain conditions cause dust explosions. Wear appropriate Personal Protective Equipment (See Section 8).

SECTION 6 – ACCIDENTAL RELEASE MEASURES

Use collection methods that minimize creation of airborne dust. High efficiency vacuuming is recommended to recover spilled dry dust. Wet product must be kept away from dry, and must not be stored in sealed containers.

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SECTION 7 – HANDLING AND STORAGE

Handling: Prevent the creation of airborne dust. Avoid ignition sources. Wear appropriate personal protective equipment. Inert atmosphere recommended when sizing to minus 20 mesh or lower.

Storage: Store in well ventilated areas. Store away from acids and bases.

SECTION 8 – EXPOSURE CONTROLS / PERSONAL PROTECTION

Airborne Exposure Limits

Based on an 8 Hour TWA:

Component

Silicon

OSHA PEL

TWA 15 mg/m³ (total)
TWA 5 mg/m³ (resp)

ACGIH TLV

TWA 10 mg/m³ (total)

Ventilation System: Local exhaust suggested in any processing areas. Ensure good ventilation.

Personal Respirators (NIOSH Approved):

Use an appropriate filtering respirator when any airborne contaminants reach or exceed the Permissible Exposure Limit (PEL). In poorly ventilated areas, such as confined spaces, a self-contained breathing apparatus or a clean air supplied respirator should be worn.

Skin Protection: Avoid any contact with skin. Wear protective gloves.

Eye Protection: Avoid any contact with eyes. Wear appropriate eye protection, and have access to eye wash stations.

Other Control Measures: Do not breathe respirable dusts.

SECTION 9 – PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Silvery grey metallic solid

Odor: None

Melting Point: 1410 °C

Boiling Point: Not available.

Freezing Point: Not available.

Solubility: Insoluble in water.

Flash Point: Not available.

Specific Gravity: 2.3

Flammability (Solid): No ignition.

pH: Not available.

Vapor Density (Air = 1): Not available.

% Volatiles by Volume @ 21 °C (70 °F): Not available.

Vapor Pressure (mm Hg): Not available.

Evaporation Rate (BuAc = 1): Not available.

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SECTION 10 – STABILITY AND REACTIVITY

Stability:	Stable under normal temperatures and dry conditions.
Conditions to Avoid:	Avoid sparks and hot work (e.g. welding and grinding etc.) in areas with high concentrations of airborne dust.
Incompatibilities:	Moisture, strong oxidizing agents, and halogen acids. Reacts rapidly with hydrofluoric and nitric acid, as well as, molten alkali.
Hazardous Polymerization:	Has not been observed.

SECTION 11 – TOXICOLOGICAL INFORMATION

Material is not known to be toxic.

SECTION 12 – ECOLOGICAL INFORMATION

Environmental Fate:	Not currently known.
Eco-toxicological Fate:	Not currently known.

SECTION 13 – DISPOSAL CONSIDERATIONS

Recovered material should be recycled, if possible.

Ensure the material is disposed of in accordance with all federal, state and local regulations.

SECTION 14 – TRANSPORT INFORMATION

DOT 49 CFR:	Not regulated for transportation.
IATA Air:	Not regulated for transportation.
IMDG Vessel:	Not regulated for transportation.
ADR:	Not regulated for transportation.

Silicon metal powder -325 mesh with the composition as described in Section 3 has been tested in accordance with the Code of Federal Regulations, Title 49, and the United Nations Transport of Dangerous Goods – Manual of Tests and Criteria, Fifth Edition (2009), and has passed the test. Therefore, silicon metal powder is not classified as a Division 4.3 dangerous when wet product.

SECTION 15 – REGULATORY INFORMATION

TSCA Listed:	Yes
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SECTION 16 – OTHER INFORMATION

NFPA Ratings - *estimated*

Health: 1
Flammability: 0
Reactivity: 1

HMIS Ratings

Health: 1
Flammability: 0
Physical Hazard: 1

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